

A Generalized Gell–Mann–Nishijima Relation for Quark and Hadron Modelling

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Abstract

The Gell–Mann–Nishijima relation is given by: $Q = T_3 + \frac{Y}{2}$. It provides the fundamental connection between electric charge Q , weak isospin T_3 , and hypercharge Y . Although it successfully reproduces the observed electric charges of all elementary particles, the physical origin of hypercharge and its relation to chirality remain implicit within the conventional formulation. Besides, it does not yield a consistent definition of hypercharge across fundamental fermions and composite particles. In this work, a generalized Gell–Mann–Nishijima relation is developed by introducing an additional internal quantum coordinate, the hyperspin projection H_3 , associated with the chiral degree of freedom. The resulting generic charge relation,

$$Q = T_3 + F + H_3,$$

where $F = (B + SN + CN + BN + TN - L)/2$ is the generalized flavour contribution constructed from the conserved matter quantum numbers, provides a single unified formula valid for all fundamental fermions and all composite hadrons without any arbitrary hypercharge assignments. The hypercharge is shown to have two physically distinct components,

$$Y = Y_F + Y_H, \quad Y_F = B + SN + CN + BN + TN - L, \quad Y_H = 2H_3,$$

where Y_F is determined by conserved matter quantum numbers and Y_H arises from the internal chiral orientation. The internal quantum state is represented by the complex hyperspin coordinate $\mathcal{H} = T_3 + iH_3$, which satisfies the invariant $T_3^2 + H_3^2 = 1/4$, providing a unified geometrical interpretation of weak isospin and chirality. The combined projection $P_3 = T_3 + H_3 = \pm \frac{1}{2}$ is shown to be chirality-independent, equalling the quark-model isospin I_3 for both handedness states of the same flavour. The formulation is validated for the complete set of Standard Model fundamental fermions across all three generations. The formalism is further extended to representative composite hadrons, including the pion triplet and nucleon isospin doublet, through the additivity of constituent quantum numbers.

Keywords: Gell–Mann–Nishijima relation; hyperspin; chirality; weak isospin; hypercharge; electroweak interactions; fundamental fermions; composite hadrons; particle quantum numbers.

1 Introduction

The classification of strongly interacting particles into symmetry multiplets has been one of the major achievements of modern particle physics. During the early 1950s, the discovery of strange particles, such as the Λ baryon and the K mesons, presented an apparent paradox: they were produced through the strong interaction on timescales of approximately 10^{-23} s but decayed through the weak interaction with lifetimes of the order of 10^{-10} s. To explain this behaviour, Nakano and Nishijima [1], and independently Gell-Mann [2], introduced strangeness S as a conserved quantum number for strong interactions. This development led to the Gell-Mann–Nishijima (GNN) relation,

$$Q = I_3 + \frac{B + S}{2} = I_3 + \frac{Y}{2}, \quad (1)$$

where Q is the electric charge, I_3 is the isospin projection, B is the baryon number, and $Y = B + S$ is the hadronic hypercharge. The relation provided the foundation for the SU(3) flavour classification known as the Eightfold Way [3, 4], culminating in the successful prediction of the Ω^- baryon and subsequently the quark model with fractional electric charges [5]. With the development of the electroweak theory, the original hadronic relation was reformulated in terms of weak isospin T_3 and weak hypercharge,

$$Q = T_3 + \frac{Y}{2}, \quad (2)$$

which forms one of the fundamental relations of the Standard Model. Although this expression reproduces the observed electric charges of all fundamental fermions, the hypercharge values for left- and right-handed fermions are assigned differently without an internal geometric explanation. For quarks, $Y = 1/3$ is assigned to both q_L doublets, while $Y = 4/3$ and $Y = -2/3$ are assigned to the singlets u_R and d_R respectively — values that are introduced as representation labels of the gauge symmetry without a common physical origin.

In this work, we propose a generalized Gell-Mann–Nishijima relation by introducing an additional internal quantum coordinate, the *hyperspin projection* H_3 , associated with the chiral degree of freedom. The generic charge relation,

$$Q = T_3 + F + H_3, \quad (3)$$

where $F = (B + SN + CN + BN + TN - L)/2$ is the generalized flavour contribution, which is valid for every particle, such as fundamental fermions and composite hadrons alike, without any additional arbitrary assignments. Comparison with Eq. (2) shows that the hypercharge decomposes as

$$Y = Y_F + Y_H, \quad Y_F = B + SN + CN + BN + TN - L, \quad Y_H = 2H_3. \quad (4)$$

The component Y_F is determined entirely by the conserved matter quantum numbers and is the same for both chiral states of the same flavour. The component $Y_H = 2H_3$ carries all the chirality dependence. The varying hypercharge assignments of the Standard Model therefore arise naturally from a single source: the chiral orientation of the internal hyperspin coordinate. The present work pursues three specific objectives:

- i. **Generic charge formulation.** A single charge relation $Q = T_3 + F + H_3$, valid for all fundamental fermions and all composite hadrons, is established in Section 2.

Comparing this with the conventional GNN equation $Q = T_3 + Y/2$ reveals that the hypercharge is not a single fundamental quantity but has two physically distinct components: $Y = Y_F + Y_H$, where Y_F is determined by the conserved baryon and lepton quantum numbers (chirality-independent) and $Y_H = 2H_3$ arises from the internal chiral orientation (chirality-dependent). This decomposition provides a consistent, non-arbitrary definition of hypercharge for every particle such that the GNN equation holds universally.

ii. **Evaluation and analysis of H_3 and hypercharge for all fundamental fermions.**

Section 3 reconstructs the hyperspin projection H_3 and the generalized hypercharge Y for all Standard Model fundamental fermions across all three generations. The results confirm that Y_F is uniform within each particle class, that $Y_H = 2H_3$ alone produces the chirality-dependent variation, and that the hyperspin invariant $T_3^2 + H_3^2 = \frac{1}{4}$ is satisfied for every state.

iii. **Extension to composite hadrons.** Section 4 demonstrates that the same generic charge relation extends to composite hadrons through the additivity of constituent quantum numbers. The net hyperspin of every externally symmetric hadron vanishes ($H_3 = 0$), recovering the classical GNN relation as a special case of the unified formula.

Section 5 discusses implications and future directions. Section 6 summarises.

2 Generalized GNN Relation in Hyperspin Space

2.1 Generic Charge Relation and Hyperspin

The Standard Model describes fermions through their electroweak quantum numbers. Left-handed fermions transform as weak-isospin doublets with $T_3 = \pm\frac{1}{2}$, whereas right-handed fermions are weak-isospin singlets with $T_3 = 0$. The conventional charge relation $Q = T_3 + Y/2$ reproduces all observed charges, but requires distinct hypercharge values to be specified separately for each chiral state.

In the present work, left- and right-handed chiral sectors are represented within a common internal quantum space by introducing an additional internal coordinate, termed *hyperspin*. The weak-isospin projection T_3 and the hyperspin projection H_3 are orthogonal components of a single internal quantum state, so that chirality becomes the orientation of the hyperspin coordinate rather than two independent quantum assignments.

Postulate 2.1 (Generic Charge Relation). *The electric charge of every particle — fundamental fermion or composite hadron — is determined by*

$$Q = T_3 + F + H_3, \quad (5)$$

where

$$F = \frac{B + SN + CN + BN + TN - L}{2} \quad (6)$$

is the generalized flavour contribution constructed from the additive conserved quantum numbers: baryon number B , strangeness SN , charmness CN , bottomness BN , topness TN , and lepton number L . The projections T_3 and H_3 are the real and imaginary parts of the complex hyperspin coordinate

$$\mathcal{H} = T_3 + iH_3, \quad (7)$$

and represent orthogonal components of a unified internal quantum state.

Equation (5) is the most general form of the charge relation in hyperspin space. For fundamental elementary fermions ($SN = CN = BN = TN = 0$) the flavour contribution reduces to

$$F = \frac{B - L}{2}, \quad (8)$$

giving the fundamental fermion charge relation

$$Q = T_3 + \frac{B - L}{2} + H_3. \quad (9)$$

For composite hadrons formed from light quarks only ($L = SN = CN = BN = TN = 0$):

$$F = \frac{B}{2}, \quad (10)$$

while for hadrons with strange content (e.g. kaons, hyperons), the full expression (6) is used, recovering the classical GNN hypercharge automatically.

2.2 Hypercharge Decomposition

Comparing the generic relation (5) with the conventional electroweak form $Q = T_3 + Y/2$, the hypercharge is seen to decompose as

$$Y = Y_F + Y_H, \quad (11)$$

where

$$Y_F = B + SN + CN + BN + TN - L \quad \text{and} \quad Y_H = 2H_3. \quad (12)$$

These two components emerge from two physically distinct origin:

Y_F is determined entirely by the conserved matter quantum numbers B , L , and the flavour quantum numbers. It is chirality-independent: for all quarks $Y_F = 1/3$ regardless of handedness or generation; for all leptons $Y_F = -1$.

$Y_H = 2H_3$ carries all the chirality dependence. For left-handed states ($H_3 = 0$): $Y_H = 0$. For right-handed up-type states ($H_3 = +\frac{1}{2}$): $Y_H = +1$. For right-handed down-type states ($H_3 = -\frac{1}{2}$): $Y_H = -1$.

The conventional hypercharge assignments, which appear to be different arbitrary numbers for each chiral state, follow from $Y = Y_F + Y_H$ with no additional freedom. Table 1 demonstrates this for first-generation quarks and leptons.

Remark 2.2. *The universal shift between left- and right-handed hypercharge is $\Delta Y = Y_H(f_R) - Y_H(f_L) = 2H_3(f_R) - 2H_3(f_L) = \pm 1$, where the sign depends only on whether the fermion is up-type or down-type. This shift is the same for all three generations and for quarks and leptons alike.*

Table 1: Hypercharge decomposition $Y = Y_F + Y_H$ for first-generation fermions. $Y_F = B-L$ is chirality-independent; $Y_H = 2H_3$ carries the full chirality dependence. The last column reproduces the Standard Model hypercharge exactly.

Particle	H_3	Y_F	$Y_H = 2H_3$	$Y = Y_F + Y_H$	Y_{SM}
u_L	0	$+\frac{1}{3}$	0	$+\frac{1}{3}$	$+\frac{1}{3}$
d_L	0	$+\frac{1}{3}$	0	$+\frac{1}{3}$	$+\frac{1}{3}$
u_R	$+\frac{1}{2}$	$+\frac{1}{3}$	+1	$+\frac{4}{3}$	$+\frac{4}{3}$
d_R	$-\frac{1}{2}$	$+\frac{1}{3}$	-1	$-\frac{2}{3}$	$-\frac{2}{3}$
ν_L	0	-1	0	-1	-1
e_L	0	-1	0	-1	-1
ν_R	$+\frac{1}{2}$	-1	+1	0	0
e_R	$-\frac{1}{2}$	-1	-1	-2	-2

2.3 Hyperspin Invariant

For the CP-symmetric fermionic states considered in this work, the hyperspin coordinate satisfies the invariant

$$|\mathcal{H}|^2 = T_3^2 + H_3^2 = \frac{1}{4}. \quad (13)$$

The invariant indicates that T_3 and H_3 are complementary projections of the same internal state. The two principal orientations are

$$(T_3, H_3) = (\pm\frac{1}{2}, 0) \quad (\text{left-handed states})$$

and

$$(T_3, H_3) = (0, \pm\frac{1}{2}) \quad (\text{right-handed states}).$$

A chirality flip corresponds to a 90° rotation in the (T_3, H_3) plane: the modulus $|\mathcal{H}| = \frac{1}{2}$ is preserved, and only the orientation changes. The hyperspin invariant therefore provides the geometrical foundation for representing LH and RH fermions as different orientations of a common internal coordinate. The phase angle

$$\theta = \tan^{-1}\left(\frac{H_3}{T_3}\right) \quad (14)$$

characterises the orientation of the hyperspin state in the (T_3, H_3) plane. For CP-symmetric states, $\theta = 0$ (LH) or $\theta = \pi/2$ (RH). CP-violating interactions may correspond to intermediate orientations where θ is dynamically significant; this extension is beyond the scope of the present work.

2.4 Chirality-Invariant Combined Projection

Proposition 2.3. *The combined projection*

$$P_3 = T_3 + H_3 \quad (15)$$

is chirality-independent. For all fundamental fermions it equals $+\frac{1}{2}$ for every up-type state and $-\frac{1}{2}$ for every down-type state, regardless of handedness.

Proof. For up-type LH states: $T_3 = +\frac{1}{2}$, $H_3 = 0$, giving $P_3 = +\frac{1}{2}$. For up-type RH states: $T_3 = 0$, $H_3 = +\frac{1}{2}$, giving $P_3 = +\frac{1}{2}$. For down-type LH states: $T_3 = -\frac{1}{2}$, $H_3 = 0$, giving $P_3 = -\frac{1}{2}$. For down-type RH states: $T_3 = 0$, $H_3 = -\frac{1}{2}$, giving $P_3 = -\frac{1}{2}$. $\square$

The projection $P_3 = T_3 + H_3$ coincides with the quark-model isospin I_3 and provides a chirality-invariant flavour label. The standard model assigns $T_3 = 0$ to RH quarks, which appears inconsistent with the quark-model statement $I_3 = \pm\frac{1}{2}$ for all quarks. The combined projection P_3 resolves this: the quark-model I_3 is not T_3 alone but the sum $T_3 + H_3$, which is the same for both chiralities.

Table 2 summarises the physical interpretation of each quantity in the generalized hyperspin formulation.

Table 2: Interpretation of quantum numbers in the generalized hyperspin formulation.

Quantity	Symbol	Interpretation
Electric charge	Q	Observable electromagnetic quantity
Flavour contribution	F	matter contribution
Weak-isospin projection	T_3	Real part of $\mathcal{H}$; gauge-visible
Hyperspin projection	H_3	Imaginary part of $\mathcal{H}$; chiral degree of freedom
Hyperspin coordinate	$\mathcal{H}$	$T_3 + iH_3$; unified internal quantum state
Chirality invariant	P_3	$T_3 + H_3 = \pm\frac{1}{2}$; equals quark-model I_3
Flavour hypercharge	Y_F	Chirality-independent hyper charge
Chiral hypercharge	Y_H	$2H_3$; carries all chirality dependence
Total hypercharge	Y	$Y_F + Y_H$; consistent for all particles

The internal quantum structure can be represented as

$$(F, \mathcal{H}) \longrightarrow (T_3, H_3, P_3) \longrightarrow (Q, Y).$$

2.5 Correspondence with the Left–Right Symmetric Representation

The complex hyperspin representation introduced in this work is fully consistent with the conventional Left–Right (LR) symmetric electroweak description. In the LR gauge group

$$SU(2)_L \otimes SU(2)_R \otimes U(1)_{B-L},$$

left- and right-handed fermions belong to independent weak-isospin spaces characterized by the generators

$$T_3^L, \quad T_3^R.$$

The hyperspin formalism combines these two orthogonal projections into a single complex internal coordinate

$$\mathcal{H} = T_3 + iH_3,$$

where

$$T_3 \equiv T_3^L, \quad H_3 \equiv T_3^R.$$

Thus, the real axis represents the conventional left-handed weak-isospin projection, while the imaginary axis represents the right-handed weak-isospin projection. Chirality therefore appears as the orientation of the hyperspin vector within a unified internal quantum space rather than as two independent quantum assignments.

With this identification, the generalized charge relation

$$Q = T_3 + \frac{B - L}{2} + H_3$$

becomes

$$Q = T_3^L + T_3^R + \frac{B - L}{2},$$

which is precisely the electric-charge operator of the Left–Right symmetric electroweak theory.

The complex hyperspin representation therefore provides a compact geometric realization of the LR electroweak structure without modifying its physical content. Throughout the remainder of this paper we employ the hyperspin notation

$$\mathcal{H} = T_3 + iH_3,$$

while recognizing its exact correspondence with the conventional Left–Right symmetry.

A detailed mathematical formulation of the complex hyperspin geometry, its algebra, and its transformation properties will be presented separately.

3 Validation for Fundamental Fermions

This section validates the generalized hyperspin formulation for all Standard Model fundamental fermions. For every fermion, the experimentally known electric charge Q , baryon number B , and lepton number L are taken as input. For elementary quarks and leptons, $SN = CN = BN = TN = 0$, so $F = (B - L)/2$. The weak-isospin assignment follows the Standard Model: $T_3 = \pm\frac{1}{2}$ for LH doublets, $T_3 = 0$ for RH singlets. The hyperspin projection is then reconstructed directly from the generic relation:

$$H_3 = Q - T_3 - \frac{B - L}{2}. \tag{16}$$

Every reconstructed state is verified against the hyperspin invariant $T_3^2 + H_3^2 = \frac{1}{4}$.

3.1 First-Generation Quarks

For quarks, $B = 1/3$, $L = 0$, giving $F = 1/6$. The charge relation is $Q = T_3 + \frac{1}{6} + H_3$.

Left-handed quarks ($H_3 = 0$, $T_3 = Q - \frac{1}{6}$):

$$u_L : T_3 = +\frac{1}{2}, H_3 = 0, \quad d_L : T_3 = -\frac{1}{2}, H_3 = 0.$$

Both left-handed quarks lie along the weak-isospin axis.

Right-handed quarks ($T_3 = 0$, $H_3 = Q - \frac{1}{6}$):

$$u_R : T_3 = 0, H_3 = +\frac{1}{2}, \quad d_R : T_3 = 0, H_3 = -\frac{1}{2}.$$

Both right-handed quarks lie along the hyperspin axis. The combined projection is $P_3 = +\frac{1}{2}$ for both u_L and u_R , and $P_3 = -\frac{1}{2}$ for both d_L and d_R , confirming Proposition 2.3.

Table 3: Reconstructed hyperspin assignments for first-generation quarks. All satisfy $T_3^2 + H_3^2 = \frac{1}{4}$ and $P_3 = T_3 + H_3 = \pm\frac{1}{2}$.

State	Q	T_3	H_3	$P_3 = T_3 + H_3$	$Y = Y_F + 2H_3$
u_L	$+\frac{2}{3}$	$+\frac{1}{2}$	0	$+\frac{1}{2}$	$+\frac{1}{3}$
d_L	$-\frac{1}{3}$	$-\frac{1}{2}$	0	$-\frac{1}{2}$	$+\frac{1}{3}$
u_R	$+\frac{2}{3}$	0	$+\frac{1}{2}$	$+\frac{1}{2}$	$+\frac{4}{3}$
d_R	$-\frac{1}{3}$	0	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{2}{3}$

3.2 All Three Generations of Fundamental Fermions

The reconstruction procedure is identical for all three quark and lepton generations. Table 4 presents the complete unified hyperspin table. The Y_F column demonstrates that the chirality-independent base hypercharge is $+\frac{1}{3}$ for all quarks and -1 for all leptons, independent of generation or handedness. The Standard Model hypercharge Y is then obtained by adding $Y_H = 2H_3$.

Several patterns are immediately visible in Table 4:

- i. The P_3 column is identical for both chiral states of every flavour, confirming Proposition 2.3 for all three generations.
- ii. The Y_F column is constant within each particle class: $+1/3$ for all quarks (all flavours, both chiralities), -1 for all leptons. This is the chirality-independent base hypercharge.
- iii. The Standard Model hypercharge differences between LH and RH states ($+1$ for up-type, -1 for down-type) are produced entirely by $Y_H = 2H_3$, confirming the decomposition (11).
- iv. All 24 entries satisfy $T_3^2 + H_3^2 = \frac{1}{4}$.
- v. The sterile right-handed neutrino ν_R emerges naturally with $Q = 0$, $T_3 = 0$, $Y = 0$ — a prediction of the hyperspin framework.

4 Extension to Composite Hadrons

Having validated the generalized formulation for all fundamental fermions, we extend it to composite hadrons. Throughout this section only externally symmetric hadronic states are considered. The generic charge relation (5) is linear in all additive quantum numbers, so the quantum numbers of a composite hadron are obtained by direct addition of the constituent-quark quantum numbers.

4.1 Additivity of the Generalized Charge Relation

For a hadron composed of constituent quarks $\{q_1, q_2, \dots\}$:

$$Q_H = \sum_i Q_i, \quad B_H = \sum_i B_i, \quad T_{3,H} = \sum_i T_{3,i}, \quad H_{3,H} = \sum_i H_{3,i}. \quad (17)$$

For antiquarks, every additive quantum number reverses sign. The net hyperspin of a composite hadron formed from constituent quarks with mixed chirality is $H_{3,H} = \sum_i H_{3,i}$.

Table 4: Unified hyperspin reconstruction for all fundamental fermions. $Y_F = B-L$ is the same for all states within each particle class. $Y = Y_F + 2H_3$ reproduces the Standard Model hypercharge exactly for every particle without arbitrary assignments.

Particle	Q	Chirality	T_3	H_3	P_3	$Y_F = B-L$	$Y = Y_F + 2H_3$
u_L	$+\frac{2}{3}$	LH	$+\frac{1}{2}$	0	$+\frac{1}{2}$	$+\frac{1}{3}$	$+\frac{1}{3}$
d_L	$-\frac{1}{3}$	LH	$-\frac{1}{2}$	0	$-\frac{1}{2}$	$+\frac{1}{3}$	$+\frac{1}{3}$
u_R	$+\frac{2}{3}$	RH	0	$+\frac{1}{2}$	$+\frac{1}{2}$	$+\frac{1}{3}$	$+\frac{4}{3}$
d_R	$-\frac{1}{3}$	RH	0	$-\frac{1}{2}$	$-\frac{1}{2}$	$+\frac{1}{3}$	$-\frac{2}{3}$
c_L	$+\frac{2}{3}$	LH	$+\frac{1}{2}$	0	$+\frac{1}{2}$	$+\frac{1}{3}$	$+\frac{1}{3}$
s_L	$-\frac{1}{3}$	LH	$-\frac{1}{2}$	0	$-\frac{1}{2}$	$+\frac{1}{3}$	$+\frac{1}{3}$
c_R	$+\frac{2}{3}$	RH	0	$+\frac{1}{2}$	$+\frac{1}{2}$	$+\frac{1}{3}$	$+\frac{4}{3}$
s_R	$-\frac{1}{3}$	RH	0	$-\frac{1}{2}$	$-\frac{1}{2}$	$+\frac{1}{3}$	$-\frac{2}{3}$
t_L	$+\frac{2}{3}$	LH	$+\frac{1}{2}$	0	$+\frac{1}{2}$	$+\frac{1}{3}$	$+\frac{1}{3}$
b_L	$-\frac{1}{3}$	LH	$-\frac{1}{2}$	0	$-\frac{1}{2}$	$+\frac{1}{3}$	$+\frac{1}{3}$
t_R	$+\frac{2}{3}$	RH	0	$+\frac{1}{2}$	$+\frac{1}{2}$	$+\frac{1}{3}$	$+\frac{4}{3}$
b_R	$-\frac{1}{3}$	RH	0	$-\frac{1}{2}$	$-\frac{1}{2}$	$+\frac{1}{3}$	$-\frac{2}{3}$
ν_{eL}	0	LH	$+\frac{1}{2}$	0	$+\frac{1}{2}$	-1	-1
e_L	-1	LH	$-\frac{1}{2}$	0	$-\frac{1}{2}$	-1	-1
ν_{eR}	0	RH	0	$+\frac{1}{2}$	$+\frac{1}{2}$	-1	0
e_R	-1	RH	0	$-\frac{1}{2}$	$-\frac{1}{2}$	-1	-2
$\nu_{\mu L}$	0	LH	$+\frac{1}{2}$	0	$+\frac{1}{2}$	-1	-1
μ_L	-1	LH	$-\frac{1}{2}$	0	$-\frac{1}{2}$	-1	-1
$\nu_{\mu R}$	0	RH	0	$+\frac{1}{2}$	$+\frac{1}{2}$	-1	0
μ_R	-1	RH	0	$-\frac{1}{2}$	$-\frac{1}{2}$	-1	-2
$\nu_{\tau L}$	0	LH	$+\frac{1}{2}$	0	$+\frac{1}{2}$	-1	-1
τ_L	-1	LH	$-\frac{1}{2}$	0	$-\frac{1}{2}$	-1	-1
$\nu_{\tau R}$	0	RH	0	$+\frac{1}{2}$	$+\frac{1}{2}$	-1	0
τ_R	-1	RH	0	$-\frac{1}{2}$	$-\frac{1}{2}$	-1	-2

Proposition 4.1. *For the externally symmetric pion and nucleon states considered here, built from left-handed quark assignments, $H_{3,H} = 0$. Consequently, the generic charge relation (5) reduces to the classical GNN formula*

$$Q_H = T_{3,H} + \frac{Y_F}{2}, \quad (18)$$

where $Y_F = B + SN + CN + BN + TN - L$ is the classical GNN hypercharge.

This demonstrates that the classical GNN relation is the $H_3 = 0$ limit of the more general hyperspin charge formula.

Mesons: The Pion Triplet: The pion isospin triplet is $(\pi^+, \pi^0, \pi^-) = (u\bar{d}, (u\bar{u} - d\bar{d})/\sqrt{2}, d\bar{u})$. Since each pion contains one quark and one antiquark, $B = 0$ and $H_{3,\pi} = 0$. The generic relation gives $Q = T_3 + 0 + 0 = T_3$:

$$\pi^+: T_3 = +\frac{1}{2} + \frac{1}{2} = +1, Q = +1; \quad \pi^0: T_3 = 0, Q = 0; \quad \pi^-: T_3 = -1, Q = -1.$$

The three pion states form the expected isospin triplet with $T_3 = (+1, 0, -1)$.

Baryons: The Nucleon Doublet: For the nucleon doublet $p = uud$ and $n = udd$, $B = 1$, $F = 1/2$, and $H_{3,N} = 0$. The generic relation gives $Q = T_3 + \frac{1}{2}$:

$$p: T_3 = +\frac{1}{2} + \frac{1}{2} - \frac{1}{2} = +\frac{1}{2}, Q = +1; \quad n: T_3 = -\frac{1}{2}, Q = 0.$$

The nucleon isospin doublet is reproduced without additional assumptions.

Table 5 summarises the reconstructed quantum numbers of representative composite hadrons.

Table 5: Quantum numbers of representative composite hadrons from the generic relation (5). All have $H_3 = 0$ so $Q = T_3 + F$. The generalised flavour contribution $F = (B + SN - L)/2$ reduces to $B/2$ for pions ($B = 0$) and nucleons ($L = SN = 0$).

Hadron	Constituents	T_3	$F = Y_F/2$	H_3	Q
π^+	$u\bar{d}$	+1	0	0	+1
π^0	$(u\bar{u} - d\bar{d})/\sqrt{2}$	0	0	0	0
π^-	$d\bar{u}$	-1	0	0	-1
p	uud	$+\frac{1}{2}$	$\frac{1}{2}$	0	+1
n	udd	$-\frac{1}{2}$	$\frac{1}{2}$	0	0

5 Discussion and Outlook

The generalized hyperspin formulation provides a single unified charge formula,

$$Q = T_3 + F + H_3, \quad F = \frac{B + SN + CN + BN + TN - L}{2},$$

that is valid for all particles without exception. The Standard Model hypercharge is not an independent fundamental quantity but decomposes naturally into two physically distinct components: Y_F , determined by conserved matter quantum numbers and chirality-independent, and $Y_H = 2H_3$, determined by the internal chiral orientation.

The practical consequence is that the “arbitrary” hypercharge assignments of the Standard Model are not arbitrary at all: they follow from two rules applied universally. (i) Every quark carries $Y_F = 1/3$; every lepton carries $Y_F = -1$. (ii) Every LH state adds $Y_H = 0$; every RH up-type state adds $Y_H = +1$; every RH down-type state adds $Y_H = -1$. These two rules, combined with $Q = T_3 + Y/2$, reproduce all known hypercharge assignments simultaneously.

The representation of the internal quantum state by the complex coordinate $\mathcal{H} = T_3 + iH_3$ with invariant modulus $|\mathcal{H}| = \frac{1}{2}$ suggests that chirality is a geometric orientation in a two-dimensional internal space. The phase angle $\theta = \tan^{-1}(H_3/T_3)$ takes the values 0 (LH) and $\pi/2$ (RH) for CP-symmetric states. CP-violating interactions may correspond to intermediate orientations; the extension of the hyperspin formalism to CP violation and quark mixing is a natural direction for future work.

The chirality-invariant combined projection $P_3 = T_3 + H_3$ reconciles the electroweak assignment $T_3 = 0$ for RH quarks with the quark-model assignment $I_3 = \pm\frac{1}{2}$ for all quarks: the quark-model I_3 is P_3 , not T_3 alone.

6 Conclusions

A generalized Gell–Mann–Nishijima relation has been developed using the hyperspin coordinate $\mathcal{H} = T_3 + iH_3$. The generic charge relation,

$$Q = T_3 + F + H_3, \quad F = \frac{B + SN + CN + BN + TN - L}{2},$$

is valid for all fundamental fermions and all composite hadrons within a single unified formula. The main conclusions are:

Universal applicability. The generic relation covers all particles. For fundamental fermions ($SN = CN = BN = TN = 0$) it reduces to $Q = T_3 + (B - L)/2 + H_3$. For composite hadrons ($H_3 = 0$) it reduces to the classical GNN relation $Q = T_3 + Y_F/2$.

Hypercharge decomposition. The Standard Model hypercharge decomposes as $Y = Y_F + Y_H$, where $Y_F = B + SN + CN + BN + TN - L$ is chirality-independent and $Y_H = 2H_3$ carries all chirality dependence. This eliminates the apparent arbitrariness of SM hypercharge assignments.

Hyperspin invariant. $T_3^2 + H_3^2 = \frac{1}{4}$ holds for all fundamental fermions across all three generations and both chiralities, making chirality a 90° rotation of the hyperspin coordinate.

Chirality invariant. $P_3 = T_3 + H_3 = \pm\frac{1}{2}$ is the same for both chiral states of every flavour, identifying P_3 with the quark-model isospin I_3 .

Validation. The formulation has been verified for all 24 chiral states of the three fermion generations (Table 4) and for representative composite hadrons (Table 5).

Classical GNN recovered. The classical GNN relation is the $H_3 = 0$ limit of the generic formula, confirming backward compatibility with all existing results.

References

- [1] T. Nakano and K. Nishijima, “Charge independence for V -particles,” *Prog. Theor. Phys.* **10**, 581 (1953).
- [2] M. Gell-Mann, “The interpretation of the new particles as displaced charge multiplets,” *Nuovo Cim.* **4** (Suppl. 2), 848 (1956).
- [3] M. Gell-Mann, “The Eightfold Way: A theory of strong interaction symmetry,” Caltech Synchrotron Laboratory Report CTSL-20 (1961).
- [4] Y. Ne’eman, “Derivation of strong interactions from a gauge invariance,” *Nucl. Phys.* **26**, 222 (1961).
- [5] M. Gell-Mann, “A schematic model of baryons and mesons,” *Phys. Lett.* **8**, 214 (1964).
- [6] S. L. Glashow, “Partial symmetries of weak interactions,” *Nucl. Phys.* **22**, 579 (1961).
- [7] P. K. Meher, *The Active Vacuum Framework: Fundamental Postulates and Physical Variables* (June 07, 2026). Available at SSRN: <https://ssrn.com/abstract=6894498>